

Sanjana Srivastava

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Data scientist with strong expertise and interest in Computer Vision, Data Mining, Large Language Models, Product Analytics, Healthcare Research, and Responsible AI.

EDUCATION

University of Virginia <i>Master of Science in Data Science</i>	Current GPA: 4.00 <i>May 2025</i>
Indian Institute of Technology, Roorkee <i>Master & Bachelor of Technology in Geological Technology, Mathematics</i>	CGPA: 8.202 <i>July 2022</i>

EXPERIENCE

Data Scientist , Deloitte (Capstone Project) - Conducting a comparative study of LLM evaluation frameworks focusing on LLM-as-a-judge vs NLP methods - Analyzing frameworks like AWS Bedrock, GCP Vertex, and RAGAS for LLM assessment - Evaluating eight metrics including accuracy, relevance, bias, hallucination, and toxicity in LLM workflows	Fall 2024 – Present
Research Scientist , University of Virginia, VA <i>Advisors - Dr. Sana Syed, Dr. Donald E. Brown</i> - Developed Deep Learning models for disease diagnosis using tissue images, clinical, and transcriptomic data - Collaborated with medical professionals to apply ML techniques for gut function analysis	2022 – 2024
Software Development Intern , BNY Mellon, India - Developed a custom XML to CSV Parser Utility and automated functional/unit testing - Conducted A/B testing to optimize internal application performance	Summer 2021
Software Development Intern , ZestMoney, India - Implemented a payment gateway using Java & SQL, improving checkout efficiency - Developed an MVVM-based Signup UI using Retrofit for Android applications	Summer 2019

PROJECTS

Evaluating efficacy of synthetic images generated using diffusion models , [Code] - Developed diffusion models to generate histology patches conditioned on nuclei locations - Validated the use of synthetic patches for improving downstream segmentation and classification tasks
Deep learning understanding of diseases using medical imaging and video data , [Code] - Implemented image cancer and gastrointestinal disease detection using Convolutional Neural Networks (CNN) - Implemented Gaussian clustering methods to identify recurring visual patterns in diseased biopsies
Humorous image captioning system , [Code] - Designed a self-attentive encoder-decoder framework to generate humorous captions for images - Developed an image-conditioned language model on a self-curated meme dataset
Correlating disease gene signature with imaging data , [Code] - Designed a deep learning framework to identify image features associated with functional gene clusters - Identified important gene signatures and their correlation with visual patterns in biopsies
Petrographic characterisation of a chondrite sample <i>Advisor - Dr. Nachiketa Rai, IIT Roorkee</i> - Investigated the mineralogy and major element geochemistry of mineral phases present in the chondrite section - Performed Electron Probe Micro Analysis to obtain backscattered electron images of the sample
Buy and sell application , [Code] <i>IIT Roorkee</i> - Developed an intranet application for buying, selling, and requesting goods among campus residents - Features include categorization of goods, subscribing, filtering, and searching based on user input

PUBLICATIONS & TALKS

- **Efficay Of Synthetic Histopathological Images In Enhancing Nuclei Segmentation Tasks**
S. Srivastava, A. Shrivastava, S. Rhoads, P.T. Fletcher, S. Syed, D.E. Brown. [Under Review]
- **What Is Normal? Characterization Of Control Pediatric Duodenal Biopsies Using Clinical Data, Machine Learning Image Analysis, And Transcriptomics**
F. Rhoads, J. Sessions, S. Srivastava, F. Zulqarnain, V. Jain, ..., S. Syed. [Poster Presentation] *ESPGHAN 2024*
- **Machine-learning-based integrative-‘omics analyses reveal immunologic and metabolic dysregulation in environmental enteric dysfunction**
F. Zulqarnain, X. Zhao, K. Setchell, Y. Sharna, P. Fernandes, S. Srivastava, ..., S. Syed. [Link] *iScience 2024*
- **What Is Normal? Characterization Of Control Pediatric Duodenal Biopsies Using Clinical Data, Machine Learning Image Analysis, And Transcriptomics**
J. Sessions, F. Rhoads, S. Srivastava, F. Zulqarnain, V. Jain, ..., S. Syed. [Oral Presentation] *DDW 2024*
- **Identifying Metabolic Signatures Of Environmental Enteric Dysfunction In Pakistani Mothers And Children Using Tissue-Specific Metabolic Modeling**
S. Syed, F. Zulqarnain, S. Srivastava, W. Khan, I. Nisar. [Poster Presentation] *Bill & Melinda Gates Foundation Grand Challenges Annual Meeting 2023*
- **What Is Normal? Characterization Of Variations In Control Duodenal Biopsies By Machine Learning Image Analysis**
F. Rhoads, J. Sessions, S. Srivastava, V. Jain, ..., S. Syed. [Poster Presentation] *NASPGHAN 2023*
- **Quantitative Morphometry and Machine Learning Model to Explore Duodenal and Rectal Mucosal Tissue of Children with Environmental Enteric Dysfunction**
M. Khan, Z. Jamil, L. Ehsan, F. Zulqarnain, S. Srivastava, ..., S. Syed. [Link] *ASTMH' 2023*

SKILLS

Languages: Python, R, SQL, C/C++, Java, JavaScript, Kotlin, L^AT_EX, Bash

Packages/Tools: PyTorch, TensorFlow, Keras, Django, React, Git, MySQL, Pandas, AWS, Hadoop, PySpark

AWARDS & ACHIEVEMENTS

- Recipient of **IIT Roorkee Award: Nayyar Award for Excellence in Communication** for winning in a multi-round competition involving debating, speech giving, essay writing, and published writing.
- Recipient of **Indian Ministry of Human Resource and Development (MHRD) Assistantship** for meeting the graduate CGPA criterion, demonstrating academic excellence exceeding the required limit.
- Cleared **Joint Entrance Examination (JEE) with All India Rank 5047 (99.6 percentile)**.

POSITIONS OF RESPONSIBILITY & EXTRA CURRICULARS

- **Teaching Assistant** for a graduate level statistical learning course at University of Virginia
- **Project Leader** at Information Management Group - Official Coding Society of IIT Roorkee
- **Editor-in-chief** at Geek Gazette - Official Technical Magazine of IIT Roorkee
- **Actor/Director/Producer** at Dramatics Section, IIT Roorkee
- **Council Member** of Hostel Management at Kasturba Bhawan, IIT Roorkee
- **Team Member** of Table Tennis National Sports Organization, IIT Roorkee